Martin C. Villanueva

Marikina City, Philippines • mcvillanueva6789@gmail.com • 0976-170-0985

Education

FEU Institute of Technology(College) — **Philippines, Manila**
Bachelor of Computer Science, Specializing in Software Engineering — **(2022-2026)**
Thesis: Deepbreath: Resnet-34 And Logistics Regression Of Cough Sounds Analysis To Assist Preliminary Diagnosis Of Pneumonia

Work and Experiences

Ollopa Corporation — **Quezon City, Philippines**
IT/E-commerce specialist (Intern) — **August 2026 – Current**

- Handled end-to-end product listings and quality assurance on Shoparino8 to maintain accurate product catalogs.
- Automated web-scraping by utilizing Python's library to get information on client's product listings and images for migration to Shoparino8.
- Migrated client inventory to Shoparino8 by web scraping client's source websites and organizing data in Google Sheets.
- Utilized AI and LLMs to automate and accelerate the creation of compelling product descriptions.

Department of Information and Communications Technology — **Quezon City, Philippines**
Quality Assurance (Intern) — **Feb 2026 – June 2026**

- Assisted in the quality assurance activities for the Regional Procurement and Bidding System, a web-based platform supporting procurement and bidding processes.
- Identified, documented, and tracked software defects, providing detailed reports to support issue resolution.
- Reviewed project documentation, business requirements, and procurement workflows to ensure comprehensive test coverage.
- Reproduced the reported bugs and simulated defects in identifying root causes.

FEU Institute of Technology — **Manila, Philippines**
Full Stack Developer(Thesis: DeepBreath) — **(2024-2025)**

- Deployed and managed the backend infrastructure on AWS EC2, hosting a Flask REST API for AI inference and application services.
- Optimized deployment, dependency management, and server configuration to support AI model execution in a cloud environment.
- Assisted in developing the Flask backend to process requests, manage user authentication, and integrate the trained ResNet-34 and Logistic Regression models.
- Developed secure cloud-to-mobile data communication using HTTP/JSON APIs for real-time inference results.

Columbia Merchandising Company — **Binondo, Manila**
IT Technical Support and Consultant (Project-Based, Contractual) — **March 2025 – December 2025**

- Set-up, prepared and configured new desktop computers and workstations.
- Installed and configured necessary software and applications.
- Provided technical support for computer and software-related issues.
- Taught employees how to use Microsoft 365 tools such as Word, Excel, Outlook, and Teams.
- Assisted with hardware and peripheral setup, including monitors, keyboards, printers, and other devices.

Website: https://portfolio-mcvillanueva-public.vercel.app

Leadership & Activities

FEU Institute of Technology
Project Coordinator(Thesis: DeepBreath)

- Coordinated technical implementation activities between the development and machine learning components of the DeepBreath project.
- Oversaw the integration of the Android application, Flask backend, cloud infrastructure, and AI inference pipeline to ensure end-to-end functionality.
- Ensured timely completion of assigned deliverables by coordinating technical tasks and supporting collaborative development efforts.

Technical Skills & Interests

Technical: Python, Git, Virtual Studio Code, Amazon Web Services(AWS), Linux, JavaScript, HTML, CSS, Zapier, Automation, n8n
Skills: Analytical Thinking, Technical Documentation, Team Collaboration
Certifications: Pearson IT-Specialist(Python), Cisco DevNet Associate, Cisco Intro to Networking, Project Management
Languages: Filipino, English
Interests: Artificial Intelligence, Machine Learning, Cloud Infrastructure and Systems, Software Engineering, Automation, Web Development, Technology and Tech Stacks

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Side Projects

Automations on Zapier
Automatic Google Form Scheduler
Workflow: Google Forms → Google Sheets → Google Calendar → Gmail

- Automated appointment scheduling using Zapier.
- Collected user information through Google Forms.
- Stored form responses in Google Sheets.
- Automatically created Google Calendar events.
- Sent appointment confirmations through Gmail.
- Reduced manual data entry and scheduling tasks.

Automations on N8N
Email Escalation
Workflow: Gmail → Local Gemma Model → Assessment → Reply/Escalation

- Built an AI-powered email automation workflow using n8n.
- Monitored incoming Gmail messages.
- Sent email content to a locally hosted Google Gemma model.
- Used the AI model to assess and classify incoming emails.
- Determined whether an email required an automated response or human intervention.
- Escalated high-priority or sensitive emails for human review.
- Demonstrated integration between LLMs and workflow automation.

HR Initial Screening Assistance using AI Workflow
Workflow: Resume Downloaded from Google Drive → Evaluation via Local AI Agent(Gemma4) → Automatic Acceptance or Rejection including the Meeting with the Email based on output of Local AI Agent(Gemma4)

- Built an AI-powered HR Initial Screening Workflow to filter applications
- Utilized Local AI Agent(Gemma4) for evaluation of Resume.
- Error Handling for multiple data entries on form submissions.
- Customized emails for each instance, rejection or continuing with a meeting.

- Automatically scheduled meetings based on the shortlisted candidates in a Google Sheet, with an additional 2-hour scheduling window for each candidate.

Basic CRM Ticketing System

SupportDesk CRM is a vibe coded full-stack customer support and relationship-management system using Google AI Studio and Google Antigravity, Visual Studio Copilot Agent. It helps organizations manage customers, track support tickets, communicate with users, monitor service activity, and control access based on user roles.

- User Authentication: Supports registration and login for administrators, support agents, and customers.
- Role-Based Access: Gives each user different permissions depending on whether they are an administrator, agent, or customer.
- Support Ticket Management: Allows users to create, search, filter, assign, update, and resolve support tickets.
- Ticket Communication: Lets customers and support staff exchange comments and view ticket activity logs.
- Analytics Dashboard: Shows ticket totals, statuses, priorities, categories, urgent issues, and recent activity.
- Customer CRM: Stores customer contact details, company information, notes, and ticket history.
- User Administration: Allows administrators to create accounts, change user roles, and delete users.
- Application Technology: Uses React and TypeScript for the frontend, Express for the backend, and a JSON file for data storage.

Demo Website: https://supportdesk-crm-mcvillanueva.vercel.app